

Z E L L I U M P R E S E N T S

MERIDIAN

RWA Tokenization Platform

Private Credit · Invoice Factoring · Trade Finance



\$50B+
RWA MARKET ON-CHAIN



3 Regions
SEA · LATAM · AFRICA



5 Modules
FULL-STACK PLATFORM

System Architecture & Flow Document

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The Opportunity

Tokenized Real-World Assets have crossed **\$50B+** on-chain (excluding stablecoins), yet the tooling to create, manage, and trade tokenized assets remains fragmented and enterprise-unfriendly.

The massive gap: Emerging market private credit — invoice factoring, trade finance, and SME lending — where traditional banks don't serve. Meridian fills this gap with a **full-stack tokenization platform** targeting traditional asset managers in Southeast Asia, LATAM, and Africa.

What Meridian Delivers



Issuer Dashboard

- KYC-gated onboarding
- SPV legal wrapper generation
- Token minting & management
- Repayment tracking



Compliance Engine

- Automated KYC/AML checks
- Jurisdiction transfer restrictions
- Tax reporting & audit trails
- OFAC/sanctions screening



Secondary Market

- Permissioned DEX (accredited only)
- Order book with depth charts
- Limit & market orders
- 0.5% platform trading fee



Yield Distribution

- Automated coupon payments
- Push & pull claim models
- Distribution calendar
- On-chain settlement



Oracle Integration

- Real-time repayment tracking from off-chain loan servicing systems
- Chainlink-compatible data feeds for asset valuations
- Automated default detection & notification triggers

Tech Stack

LAYER	TECHNOLOGY	PURPOSE
Blockchain	Ethereum + Avalanche	Institutional trust + permissioned subnets
Token Standard	ERC-3643 (T-REX)	Compliant security tokens with identity hooks
Identity	On-chain DID + VCs	Verifiable Credentials for KYC/accreditation
Legal Layer	SPV Wrappers	Centrifuge-style legal entity structure
Frontend	Vite + React 18	Premium dashboard with ethers.js v6
Contracts	Hardhat + Solidity 0.8	6 core smart contracts
Backend	Node.js + Express	API layer, oracle service, KYC mock
Charts	Recharts	Portfolio & yield data visualization

Smart Contract Architecture



Flow 1 — Asset Issuer Journey

SMEs, lenders, and originators tokenizing private credit instruments

01

Onboarding & KYC

Issuer registers on Meridian, submits entity docs (incorporation, tax ID, director KYC). Compliance Engine runs automated checks against OFAC, PEP lists, and jurisdiction rules.

**02**

SPV Creation

Platform generates a legal wrapper (Special Purpose Vehicle) for the asset pool. SPV isolates credit risk from the originator. Jurisdiction selected (Cayman, BVI, Singapore).

**03**

Asset Configuration

Issuer uploads invoice/loan details: debtor info, face value, maturity date, interest rate, risk rating. Documents stored in encrypted vault.

**04**

Token Minting

CreditToken.sol deploys an ERC-3643 compliant token representing the credit instrument. Transfer restrictions are encoded on-chain via ComplianceModule.

**05**

Marketplace Listing

Token appears on the Investor Marketplace. Accredited investors can browse, analyze risk metrics, and invest. Oracle begins tracking repayments.



06

Yield & Maturity

YieldDistributor.sol automates coupon payments to token holders. At maturity, tokens are redeemed (burned) and principal returned to investors.

Flow 2 — Investor Journey

Accredited investors deploying capital into tokenized private credit

01

Wallet Connect & KYC

Investor connects MetaMask/WalletConnect. Completes KYC verification — proof of identity, accreditation status, and jurisdiction. On-chain DID credential issued.



02

Browse Marketplace

Explore available credit instruments with filters: yield range, risk rating, maturity, jurisdiction. Each listing shows SPV details, debtor profile, and projected cash flows.



03

Due Diligence

Deep-dive into asset details: legal documents, credit scores, repayment history, oracle data feeds. Investment calculator shows projected returns.



04

Invest (Primary Market)

Investor commits USDC/USDT to purchase credit tokens. Smart contract verifies identity compliance before executing the transfer. Tokens appear in portfolio.



05

Trade (Secondary Market)

Investor can list tokens on the Permissioned DEX. Only verified investors can place buy/sell orders. Order book matching with 0.5% platform fee.



06

Yield Collection

Automated coupon payments distributed to wallet. Investor tracks yield on the dashboard. At maturity, principal is returned and tokens are burned.

Flow 3 — Compliance & Transfer Logic



Every Token Transfer Passes Through This Pipeline

The ERC-3643 standard enforces compliance at the smart contract level. No transfer can occur without passing all checks — this is enforced on-chain and cannot be bypassed.

①

Identity Check

IdentityRegistry verifies both sender and receiver have valid on-chain DIDs with active KYC credentials.

②

Jurisdiction Check

ComplianceModule validates that the transfer doesn't violate country-to-country restrictions (e.g., US → sanctioned nations).

③

Investor Type Check

Verifies receiver is accredited / qualified for the specific asset class and meets minimum investment thresholds.

④

Holding Period Check

Enforces lock-up periods — tokens cannot be transferred before the minimum holding duration expires.

⑤

Max Holders Check

Ensures the token doesn't exceed the maximum number of holders (regulatory cap for certain security types).

⑥

Transfer Executes ✓

All checks pass → token transfer is executed atomically on-chain. Full audit trail recorded.

Forced Transfer (Regulatory Override)

In cases of court orders, lost keys, or regulatory action, authorized agents can execute **forced transfers** — moving tokens between wallets without standard compliance checks. This is a critical feature for regulated securities that standard ERC-20 tokens lack.

Revenue Streams

REVENUE STREAM	RATE	DESCRIPTION
Issuance Fee	0.5 — 1%	One-time fee on total token issuance volume
Management Fee	0.25% / yr	Annual fee on assets under management
Trading Fee	0.5%	Per-trade fee on secondary market transactions
Compliance SaaS	Subscription	Monthly compliance-as-a-service for issuers

Target Market

 **Southeast Asia**
Invoice factoring for SMEs
Trade finance corridors
Malaysia, Singapore, Philippines

 **LATAM**
SME lending markets
Cross-border trade finance
Brazil, Mexico, Colombia

 **Africa**
Agricultural trade finance
Microfinance tokenization
Nigeria, Kenya, South Africa

Competitive Advantage

Centrifuge, Maple, and Goldfinch focus on DeFi-native users. Meridian targets **traditional asset managers** in emerging markets who want to tokenize but need hand-holding. Our platform abstracts blockchain complexity behind familiar financial workflows — no crypto expertise required.